

Martin Orozco Jr.

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OBJECTIVE

Electrical Engineering student at San José State University seeking a Summer 2026 internship in power electronics and hardware applications, applying experience in circuit analysis, PCB design, and system-level troubleshooting.

EDUCATION

- Bachelor's of Science, Electrical Engineering** Expected - Dec. 2027
San Jose State University; GPA: 3.75 San Jose, CA
- Electrical Engineering Transfer Degree** May 2025
Mission College; GPA: 3.75 Santa Clara, CA

EXPERIENCE

- SJSU Formula SAE Spartan Racing** Oct 2025 - Present
Electrical engineering Intern, Sensor and Harness Team San Jose, CA
 - Sensor Integration and Validation:** Supported development and validation of vehicle sensor systems, defining electrical integration requirements and verifying signal reliability
 - Sensor/Harness Verification:** Supported subsystem integration testing and pre-validation checks prior to vehicle operation and competition
 - Cross-Team Collaboration:** Coordinated with mechanical and battery teams to ensure reliable sensor placement and harness routing under vibration and thermal constraints
- Foreman/Master ASE Technician** Nov 2011 - Aug 2025
BMW of Fremont Fremont, CA
 - Technical Leadership:** Supervised and supported a team of 24 technicians in diagnosing complex electrical and mechanical faults, ensuring repair accuracy and efficiency
 - High-Voltage Systems Expertise:** Performed diagnostics and repairs on BMW hybrid and electric vehicle systems, following high-voltage safety protocols and isolation procedures
 - Electrical Diagnostics:** Utilized ISTA diagnostic software, oscilloscopes, and wiring diagrams to troubleshoot CAN communication faults and analyze digital signal behaviors
 - Root Cause Analysis:** Identified intermittent electrical faults through systematic signal tracing, load testing, and component validation under real-world operating conditions
 - Process Optimization:** Improved shop workflow and repair turnaround efficiency, coordinating job prioritization, quality control inspections, maintaining peak productivity and quality standards
 - Quality and Compliance:** Ensured repairs met BMW factory standards and safety regulations, maintaining high customer satisfaction and reduced comeback rates
 - Mentorship and Training:** Trained junior technicians on diagnostic strategy, electrical theory application, and safe handling of high-voltage systems

PROJECTS

- Window Comparator PCB Design:** Designed and simulated a window comparator circuit in LTspice and developed a fabrication-ready PCB in Altium Designer, generating schematics, Gerber files, and BOM

TECHNICAL SKILLS

Programming: C++, Python, Verilog, MATLAB

Hardware & Electrical: Analog/Digital Circuit Analysis, Sensor Integration, Signal Conditioning, Wiring Harness Design, PCB Design, High-Voltage Safety Practices

Test & Measurement: Oscilloscopes, Multimeters, Power Supplies, Signal Debugging, Hardware Validation

Tools & Software: Altium Designer, LTspice, MATLAB, Vivado, RapidHarness

Systems & Interfaces: CAN Bus Diagnostics, Embedded Systems Fundamentals, FPGA Development

Fabrication & Assembly: Soldering, Rework, Connector Crimping, Harness Assembly